Spatial–Frequency Deepfake Image Classification with Wavelet-Packet Features and Bidirectional Cross-Attention

Abstract

Reliable deepfake image detection under lossy compression and generator shift remains challenging. This study describes an image-level binary classifier combining an ImageNet-pretrained Xception spatial branch with a differentiable three-level Daubechies-4 wavelet-packet decomposition (WPD) branch, a learnable high-frequency enhancement (HiFE) module, and bidirectional dual cross-attention (DCA) fusion. The image-level experiment uses a 140,000-image real-versus-synthetic face collection identified as FFHQ versus StyleGAN2, partitioned into 100,000 training, 20,000 validation, and 20,000 test images with balanced classes. The reported ROC-AUC point estimates are 0.937 at JPEG quality 50 and 0.888 at JPEG quality 23. Implementation testing reports 13 successful architecture tests and six successful differentiable-WPD tests on an NVIDIA RTX 4060 Laptop GPU using PyTorch 2.6.0 and CUDA 12.4. Matched per-variant ablation metrics, repeated-seed uncertainty estimates, numeric cross-generator results, and native FaceForensics++ video results are not reported here. The findings therefore characterize initial JPEG behavior on one image collection and do not establish independent gains from WPD, HiFE, or DCA, or generalization to video domains.

Keywords: deepfake detection; digital image forensics; wavelet-packet decomposition; JPEG robustness; spatial-frequency fusion; cross-generator evaluation; reproducibility.

1 Introduction

The increasing realism of manipulated and synthesized facial media creates challenges for digital trust, identity verification, and forensic analysis. Deep-learning detectors have been developed to identify such content from spatial appearance, frequency statistics, temporal inconsistencies, and combinations of these cues. FaceForensics++ established a widely used benchmark for manipulated facial videos and highlighted the importance of evaluating detectors under different compression conditions [1]. Xception-based convolutional detectors are also common baselines for facial manipulation detection, in part because their depthwise-separable convolutions provide expressive spatial features at a practical computational cost [2].

A key challenge is that performance on a held-out split from one dataset does not necessarily imply robustness to changes in generator, source domain, codec, or post-processing. Frequency-domain methods exploit artifacts that may be difficult to identify from RGB appearance alone. Frank *et al.* investigated frequency analysis for recognizing GAN-generated images, while Qian *et al.* introduced a frequency-aware framework for face forgery detection [3, 4]. However, frequency features can also encode generator-specific or dataset-specific shortcuts; recent work on frequency debiasing explicitly examines this concern [5].

Wavelet-packet decomposition has previously been studied for deepfake image analysis [6]; wavelet-packet filter-bank methods are grounded in multiresolution and best-basis analysis [7, 8]. Later research has explored content-guided spatial-frequency reasoning, adaptive fusion, and high-frequency enhancement for compressed content [9–11]. Therefore, this paper does not claim

that WPD, frequency-domain analysis, or adaptive spatial-frequency fusion is novel by itself. Instead, it asks whether a particular combination of a three-level db4 WPD representation, a learnable enhancement block, and cross-attention fusion provides a measurable incremental benefit under controlled training conditions.

The central research question is:

> *Under matched data splits and optimization budgets, does the proposed WPD-enhanced spatio-frequency detector improve detection under compression and domain shift beyond the benefits attributable to compression-aware augmentation alone?*

The study is organized around five testable hypotheses:

- **H1:** Compression-aware augmentation improves detection under the tested degradation conditions.
- **H2:** WPD contributes an incremental benefit over a spatial-only baseline when training conditions are matched.
- **H3:** HiFE contributes measurable benefit beyond the same model without enhancement.
- **H4:** DCA provides measurable benefit over simple concatenation under a comparable training protocol.
- **H5:** Any observed in-domain improvements persist, at least partly, under external-dataset or unseen-generator evaluation.

These hypotheses are empirical. None is considered supported until the corresponding experiment has been run and its uncertainty reported.

The paper makes the following intended contributions. First, it specifies a reproducible dual-branch design in which spatial appearance and wavelet-packet features are processed separately and then fused. Second, it defines matched component ablations to isolate the effects of WPD, HiFE, DCA, and compression-aware augmentation. Third, it establishes an evaluation protocol that separates native FaceForensics++ video compression from additional JPEG transformations and includes external-domain testing. Fourth, it requires reproducible data manifests, repeated seeds, calibrated operating points, and test procedures whose outputs are derived from actual experimental runs.

The completed empirical scope is limited to the 140k real/fake face collection identified as FFHQ versus StyleGAN2 and evaluated under JPEG image recompression. Two quantitative AUC values are available; the GAN-to-diffusion transfer experiment is described only qualitatively and lacks sample counts, generator details, and uncertainty estimates. Native FaceForensics++ video evaluation and complete component-wise ablations remain outstanding. The attention module operates at the spatial-frequency fusion stage rather than applying attention directly to individual wavelet packets.

2 Related Work

2.1 Spatial-Domain Detection

Spatial detectors learn discriminative patterns from RGB pixels. Xception uses depthwise-separable convolutions to model spatial and channel relationships efficiently [2]. The Face-

Forensics++ benchmark provides a common setting for evaluating facial manipulation detectors under different compression conditions [1]. While spatial representations are strong baselines, their results should be interpreted in the context of the dataset, manipulation methods, and compression pipeline used for evaluation. Generalization must be measured rather than inferred from within-dataset performance.

2.2 Frequency-Domain Forensics

Frequency representations can reveal irregularities associated with image synthesis and upsampling. Frank *et al.* studied frequency analysis for deep fake image recognition [3]. Qian *et al.* proposed F^3-Net, which combines frequency-aware decomposed image components with local frequency statistics [4]. These studies demonstrate that frequency information is an established direction in forgery analysis; consequently, using a frequency transform alone is not a sufficient novelty claim.

2.3 Wavelets and Wavelet-Packet Decomposition

Wolter *et al.* introduced wavelet-packet analysis for deepfake image detection and examined its use for spatial-frequency forensic features [6]. A conventional discrete wavelet transform recursively decomposes the low-frequency approximation branch, while a wavelet-packet transform recursively decomposes low- and high-frequency nodes. This provides a more complete multiresolution partition of the signal, but the usefulness of a representation still depends on the task, implementation, and evaluation protocol. This work investigates a differentiable db4 WPD implementation as a component of a larger detector; it does not claim to introduce WPD for deepfake detection.

2.4 Spatial-Frequency Fusion and High-Frequency Enhancement

Research has explored interaction between spatial content and frequency clues. Wang *et al.* proposed content-guided spatial-frequency relation reasoning with dynamic graph learning [9]. Wang *et al.* later investigated adaptive fusion of spatial-frequency features [10]. Gao *et al.* studied a high-frequency enhancement network for highly compressed deepfake content [11]. These works overlap conceptually with the motivation for combining spatial features, frequency information, and learnable feature weighting. The present study therefore needs controlled experiments to establish what, if anything, is added by its particular combination of WPD, HiFE, and DCA.

2.5 Generalization and In-the-Wild Evaluation

FreqDebias identifies the risk that deepfake detectors may over-rely on selected frequency bands associated with particular forgery types, motivating explicit analysis of spectral bias and cross-domain performance [5]. The 2026 WGN work also investigates wavelet-guided spatial attention, compression, and external-dataset generalization [12]. Deepfake-Eval-2024 provides a recent in-the-wild benchmark assembled from media circulated in 2024, illustrating the importance of moving beyond academic test conditions [13]. These studies raise the bar for a convincing evaluation: testing only one real-image source, one synthetic generator, or a few quality settings is insufficient for broad deployment claims.

2.6 Research Gap

Prior work has separately established the relevance of spatial detectors, frequency-based cues, WPD, feature fusion, and high-frequency enhancement. The remaining question for this project is narrower: whether the selected combination provides benefits that remain after controlling for augmentation, training budget, and data distribution. This is particularly important because a full model trained with more data or more optimization steps cannot be fairly compared with a smaller baseline trained under a weaker protocol. The study therefore prioritizes matched ablations, dense compression evaluation, external-domain testing, and uncertainty estimates over a claim of novelty based only on the architecture.

3 Problem Formulation

Let an input facial image or extracted video frame be represented by $X \in [0,1]^{H \times W \times 3}$. A sample may have undergone an unknown sequence of compression or post-processing transformations. The detector estimates

$$p_\theta(y = 1 \mid X) = \sigma(f_\theta(X)), \tag{1}$$

where $y = 1$ denotes manipulated content, $y = 0$ denotes authentic content, f_θ is the model logit, and $\sigma(\cdot)$ is the sigmoid function. The task is image-level binary classification. The implementation has no segmentation decoder or pixel-mask supervision; consequently, Grad-CAM or attention maps are used only for qualitative feature inspection and are not interpreted as pixel-level localization measurements.

The goal is not merely to maximize performance on one test split, but to quantify changes across source compression conditions and independent datasets. The primary comparisons should use the same source samples, preprocessing, training budget, and validation-based model selection wherever possible.

4 Proposed Method

4.1 Overview

The proposed framework contains two branches. The spatial stream encodes RGB appearance using Xception features [2]. The frequency stream converts the image to luminance, applies a three-level db4 WPD, and processes the resulting packet maps with a learnable enhancement block. A dual cross-attention module then allows the two streams to exchange information before binary classification. Fig. 1 summarizes the data flow.

4.2 Face Acquisition and Preprocessing

When the source is a video, frames are extracted using a documented and fixed sampling policy. Face regions are then detected using a documented face detector such as MTCNN or RetinaFace. The specified preprocessing pipeline uses a 20% bounding-box margin to retain blending boundaries, five-landmark alignment when available, and bilinear resizing to 224×224 pixels. If alignment fails, the fallback policy and failure rate must be reported; frames must not be silently dropped in a class-dependent manner.

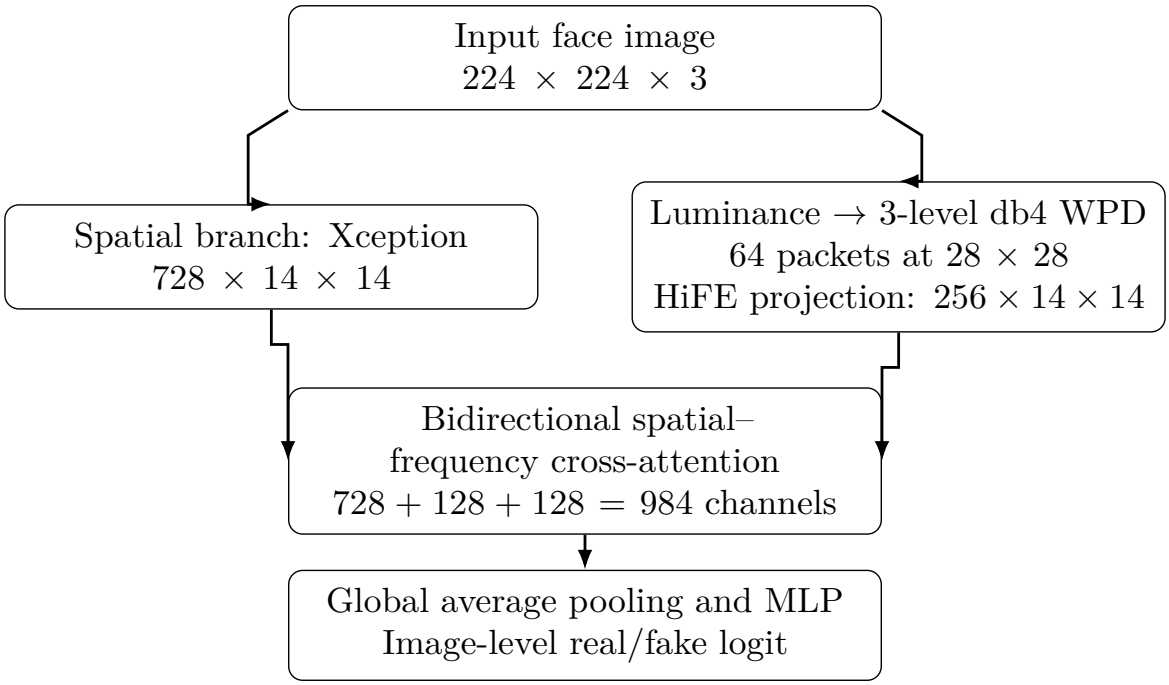


Figure 1: Single-image processing pipeline. The three-level WPD branch produces 64 packet maps; HiFE projects its output to the 14×14 fusion grid. DCA exchanges information between spatial and frequency tokens, and the classifier outputs one image-level logit.

Let $X_{\text{rgb}} \in [0,1]^{224\times224\times3}$ be the resized RGB crop. Branch-specific normalization is applied after the luminance signal is calculated, avoiding ambiguity between normalized RGB values and the luminance coefficients. The same decoding, resizing, and crop policy must be applied to authentic and manipulated samples.

4.3 Spatial Representation

The spatial branch uses an ImageNet-1k-pretrained Xception implementation accessed through `timm`, with a feature tap reported at Middle Flow Stage 3. For an input batch $X_{\text{rgb}} \in \mathbb{R}^{B\times3\times224\times224}$, the reported implementation output is

$$F_s = G_s(X_{\text{rgb}}; \theta_s) \in \mathbb{R}^{B\times728\times14\times14}. \tag{2}$$

The implementation report identifies this as the actual tap point, rather than a dimension obtained by arbitrary reshaping. Reproduction should record the precise `timm` version and weight identifier used for the run.

4.4 Luminance Wavelet-Packet Branch

For RGB values represented in the $[0,1]$ range, luminance is calculated as

$$Y = 0.299R + 0.587G + 0.114B. \tag{3}$$

The differentiable two-dimensional wavelet-packet transform uses fixed db4 quadrature-mirror filter banks of length $K = 8$. The separable filters are formed from low- and high-pass filters as $LL = h_L \otimes h_L$, $LH = h_L \otimes h_H$, $HL = h_H \otimes h_L$, and $HH = h_H \otimes h_H$. At each of three levels

every packet is decomposed, resulting in $4^3 = 64$ terminal bands. The implementation report specifies reflect padding $(3, 4, 3, 4)$ before stride-2 convolution at each level, giving the dyadic size transitions $224 \to 112 \to 56 \to 28$. Thus,

$$W_3 = \mathrm{WPD}_{\mathrm{db4},L=3}(Y) \in \mathbb{R}^{B\times 64\times 28\times 28}. \tag{4}$$

The reported test suite checks filter-bank properties and gradient propagation. The implementation supports both log-magnitude, $\log(|W_3| + \epsilon)$, and log-energy, $\log(W_3^2 + \epsilon)$, representations. The exact representation mode used to produce the AUC values must be recorded from the corresponding experiment configuration before publication; these definitions should not be treated as interchangeable.

4.5 High-Frequency Enhancement

The enhancement module is designed to learn channel-wise weighting and local transformations of WPD packet features. Let W denote the stacked packet tensor and let z_{avg} and $z_{\max}$ be global average- and max-pooled channel descriptors. A channel gate can be expressed as

$$g = \sigma\left(\mathrm{MLP}_a(z_{\mathrm{avg}}) + \mathrm{MLP}_m(z_{\max})\right), \tag{5}$$

followed by a residual feature transformation

$$W_e = W \odot g + \Phi(W), \tag{6}$$

where Φ is a learned convolutional transformation and $\odot$ denotes element-wise multiplication with appropriate broadcasting. The exact convolution type, kernel sizes, normalization, and activation functions must match the source implementation and be specified in the final version. Because compressed media may have lost information, this module should be described as feature enhancement or reweighting unless a separate reconstruction objective demonstrably supports stronger reconstruction claims.

The enhanced frequency tensor is projected to the fusion resolution. The intended projected representation is

$$F_f \in \mathbb{R}^{256\times 14\times 14}. \tag{7}$$

The projection operation and its output dimensions require a model-shape test.

4.6 Dual Cross-Attention Fusion

The spatial and frequency features are aligned on a 14×14 grid. Flattening this grid gives $N = 196$ tokens per branch. The reported DCA implementation projects each direction to a 128-dimensional attended representation. Let $F_s' \in \mathbb{R}^{B\times 196\times 728}$ and $F_f' \in \mathbb{R}^{B\times 196\times 256}$ denote the spatial and frequency token sequences. Query, key, and value projections are learned so that each value sequence has 128 channels:

$$Q_s = F_s' W_{Q_s}, \qquad K_f = F_f' W_{K_f}, \qquad V_f = F_f' W_{V_f}, \tag{8}$$

$$Q_f = F_f' W_{Q_f}, \qquad K_s = F_s' W_{K_s}, \qquad V_s = F_s' W_{V_s}. \tag{9}$$

Following the general scaled dot-product attention formulation [14], the bidirectional cross-attention outputs are

$$A_{s\to f} = \mathrm{Softmax}\left(\frac{Q_s K_f^\top}{\sqrt{d_k}}\right) V_f, \tag{10}$$

$$A_{f\to s} = \mathrm{Softmax}\left(\frac{Q_f K_s^\top}{\sqrt{d_k}}\right) V_s, \tag{11}$$

where the query/key projection dimension is d_k and each attended output has shape $B \times 196 \times 128$. Reshaping the outputs back to feature maps and concatenating them with the spatial bypass yields

$$F_{\text{fused}} = [F_s \| A_{s\to f} \| A_{f\to s}] \in \mathbb{R}^{B\times 984\times 14\times 14}, \tag{12}$$

since $728 + 128 + 128 = 984$. This is the channel accounting reported for the implementation. It should not be confused with concatenating the 256-channel projected frequency tensor directly to the spatial stream. Any exact replication should additionally record the implemented attention head count, normalization, and projection details from the model configuration.

4.7 Classification and Optimization

Global average pooling maps the fused feature tensor to a vector $z \in \mathbb{R}^{984}$. The proposed classification head applies a fully connected layer with 512 units, batch normalization, ReLU, dropout with $p = 0.5$, and a final linear projection to a scalar logit ℓ. The manipulated-class probability is $\hat{p} = \sigma(\ell)$.

For binary labels $y_i \in \{0, 1\}$ and smoothed labels

$$y_i^* = y_i(1 - \epsilon) + \frac{\epsilon}{2}, \qquad \epsilon = 0.05, \tag{13}$$

the binary cross-entropy objective is

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} \left[y_i^* \log \hat{p}_i + (1 - y_i^*) \log(1 - \hat{p}_i)\right]. \tag{14}$$

The implementation should use a numerically stable loss, such as a logits-based binary cross-entropy implementation, when the output supplied to the loss is a logit. The sigmoid should then be applied only for probability reporting and threshold-based decisions.

5 Experimental Protocol

5.1 Datasets and Evaluation Roles

The image-level experiments use the "140k Real and Fake Faces" collection, identified in the project dataset configuration as FFHQ real images versus StyleGAN2-generated images. StyleGAN2 is described by Karras *et al.* [15]. The dataset is reported to contain 140,000 balanced images partitioned into 100,000 training images (50,000 real and 50,000 fake), 20,000 validation images (10,000 real and 10,000 fake), and 20,000 test images (10,000 real and 10,000 fake). All AUC values reported here refer to this static-image collection, not to FaceForensics++ video.

FaceForensics++ is a planned video benchmark. The experiment workspace used for this report contained preparation scripts and directory scaffolding but not the video data; thus, no FF++ C0/C23/C40 results are reported. Celeb-DF v2 and DFDC are intended external benchmarks [16,17], but completed metrics for these datasets are not available. Table 1 distinguishes completed image-level evaluation from planned video and external-domain testing.

Table 1: Dataset roles and current evidence status.

Dataset	Role	Current status
140k Real/Fake Faces	Completed image-level JPEG evaluation; described as FFHQ versus StyleGAN2	Reported split: 100k train, 20k validation, 20k test; each split is balanced.
FF++	Planned video benchmark under native C0/C23/C40 conditions	Videos not present in the reported workspace; no results claimed here.
Celeb-DF v2 and DFDC	Intended external-domain benchmarks	Do not report completed metrics unless the experiment output and split metadata are available.

5.2 Partitioning and Leakage Prevention

Split video/source identities before extracting frames. No source video or near-duplicate frame should appear in more than one partition. Perform exact-hash checks and perceptual-hash checks where appropriate. Audit whether real and fake classes differ in metadata, resolution, crop distribution, encoding, or background in ways that could become shortcuts. Generate train, validation, and test manifests once, version them, and record their hashes in every experiment log.

5.3 Compression Protocol

The completed image-level evaluation reported in the available experiment record uses JPEG quality factors $Q \in \{100, 50, 23\}$ and a broader sweep $Q \in \{100, 90, 75, 50, 30, 20, 10\}$. The reported quantitative points currently available are for JPEG $Q = 50$ and $Q = 23$; the quality-10 result is described only as a decline without a numeric AUC. JPEG quality factors describe image recompression and must remain distinct from FaceForensics++ video compression labels C0, C23, and C40. The native FF++ video evaluation was not completed in the reported workspace and is not presented as an experimental result here.

5.4 Matched Ablations

The implementation report describes six configurable model variants in `models/team14_net.py`. Their existence establishes that the code supports the comparisons; it does not establish that all six variants have completed matched training and evaluation. The variants are listed in Table 2. A fair comparison must match the data split, preprocessing, optimization budget, model selection, and JPEG transformations as closely as possible.

5.5 Training Configuration and Verification Environment

The architecture verification environment was an NVIDIA GeForce RTX 4060 Laptop GPU with 8 GB VRAM, CUDA 12.4, and PyTorch 2.6.0+cu124. The initial development specification listed

Table 2: Six implemented architecture variants and the component each is intended to isolate.

ID	Variant key	Purpose
A	`spatial_only`	Xception spatial baseline.
B	`wpd_only`	Frequency-branch baseline.
C	`spatial_wpd_concat`	Measures the effect of adding WPD with simple concatenation.
D	`spatial_wpd_hife`	Evaluates enhancement with the spatial-frequency model.
E	`spatial_wpd_dca`	Evaluates DCA without HiFE.
F	`full_team14`	Full WPD + HiFE + DCA system.

PyTorch 2.2 and CUDA 12.1; the environment above is the one associated with the reported implementation tests. The run-level configuration linked to each AUC value should include the optimizer, batch size, number of epochs or optimization steps, augmentation settings, random seed, and WPD representation mode. These fields were not all present in the available result record and therefore are not inferred here.

The standardized protocol for future matched ablations specifies AdamW, a batch size of 32, 20 epochs, two warm-up epochs, a cosine learning-rate schedule, a 5.0×10^{-5} learning rate for the spatial backbone and 1.0×10^{-4} for the wavelet/fusion/head parameters, and weight decay 10^{-4}. It prescribes horizontal flipping, mild color jitter, Gaussian blur, and JPEG augmentation with $Q \in [60, 100]$ on the training split only. These settings define the reproducible ablation protocol; the available run-level records do not establish that each setting exactly matches the training run that produced the two reported AUC points.

The project protocol specifies five seeds, $s \in \{0, 1, 2, 3, 4\}$, for principal comparisons where resources permit. However, the available experiment record does not identify the number of seeds used for the two quoted AUC values. Those values are therefore reported as point estimates, not as five-seed means. Repeated-seed mean, standard deviation, confidence intervals, and paired statistical comparisons remain required for the completed ablation study.

5.6 Metrics and Threshold Selection

Report ROC-AUC, accuracy, balanced accuracy, precision, recall, F1-score, confusion matrices, false-positive and false-negative rates, and calibration measures where relevant. Equal Error Rate may be included if its calculation and interpolation procedure are explicitly specified. Define the number and strategy of ECE bins and provide reliability diagrams alongside the Brier score.

Select the operating threshold on the validation set only and freeze it before test evaluation. Report performance at 0.5 separately where useful. Confidence intervals and hypothesis tests should account for the unit of dependence; correlated frames from the same video must not be treated as independent observations. Use a paired bootstrap or DeLong test for AUC comparisons when appropriate, and report effect sizes, confidence intervals, and corrections for multiple comparisons.

5.7 Efficiency and Reproducibility

Measure throughput, per-image or per-frame latency, and peak memory separately. For GPU timing, perform warm-up iterations and synchronize CUDA before and after the timed region.

Compare methods under equivalent batch sizes and input shapes, and report repeated timings. Save model checkpoints, configs, package versions, dataset-manifest identifiers, seeds, and evaluation outputs. Figures and tables must be generated from stored results rather than hardcoded values.

6 Results

The reported result set contains two quantitative JPEG AUC values but does not include the run count, confidence intervals, or all machine-readable evaluation artifacts needed to establish whether these values are single-run measurements or aggregates. They are therefore reported as point estimates, not as multi-seed averages. No missing measurements are inferred.

6.1 JPEG Compression Evaluation

The available image-level findings are summarized in Table 3. On the FFHQ-versus-StyleGAN2 static-image test split, the reported AUC is 0.937 at JPEG $Q = 50$ and 0.888 at JPEG $Q = 23$. Performance is described as lower at $Q = 10$, but no numeric AUC is available for that condition.

Table 3: Reported JPEG-quality results on the FFHQ-versus-StyleGAN2 image evaluation. Values are point estimates from the available experiment record; seed aggregation and confidence intervals were not supplied.

JPEG quality Q	ROC-AUC	Evidence status
50	0.937	Numeric value reported in the available experiment record.
23	0.888	Numeric value reported in the available experiment record.
10	Not supplied	A decline is reported, but the numeric AUC is unavailable.
100	Not supplied here	Included among evaluated quality factors, but no numeric AUC was supplied.

These results indicate that the reported detector retains discriminative ability at the two stated JPEG settings, with lower AUC at $Q = 23$ than at $Q = 50$. They do not establish resilience to FF++ C40, since JPEG image quality and FF++ video compression levels are different protocols. Nor do these two points alone define a full robustness curve or demonstrate that WPD, HiFE, or DCA independently caused the observed performance.

6.2 Implementation Verification

The implementation verification outcomes are summarized in Table 4. The reported tests assess the model components and differentiable filter-bank implementation in the stated CUDA environment. Unit tests provide implementation-level evidence; they do not, by themselves, establish comparative detector accuracy or generalization.

Table 4: Implementation verification results reported by the project.

Test suite	Passed	Reported runtime
`test_team14_architecture.py`	13/13	3.344 s on CUDA.
`test_differentiable_wpd.py`	6/6	0.732 s on CUDA.

The implemented WPD filter bank uses fixed registered buffers and supports input gradients. Under the specified reflect-padding and dyadic-downsampling convention, it produces 64 packet maps of 28×28 from a 224×224 luminance image. The mathematical tests are reported to check unit filter energy, pairwise orthogonality, and finite non-zero gradients propagated to the input. The reported feature shapes are $728 \times 14 \times 14$ for the spatial stream, $256 \times 14 \times 14$ after HiFE, and $984 \times 14 \times 14$ after fusion. Reproducibility additionally requires dependency versions and run-specific configuration records.

6.3 Ablation and Cross-Generator Evidence

The six canonical architecture variants are implemented in the repository, but the available experiment record does not contain their full per-variant metrics under a matched protocol. Consequently, no ranking among the spatial-only, WPD-only, concatenation, HiFE, DCA, and full-model variants is reported here. In particular, the available information does not establish that DCA outperforms simple concatenation or that every added component contributes an independent gain.

A qualitative drop is reported when transferring from GAN-generated content to diffusion-generated content. However, the transfer dataset counts, generator list, numeric AUC, and uncertainty estimates are not specified, so the magnitude of this domain shift cannot be quantified from the current results. Cross-generator robustness therefore remains an open evaluation question.

6.4 Planned External Evaluation

No FaceForensics++ video files were available in the evaluated workspace, so no FF++ C0/C23/C40 metrics are reported. Celeb-DF v2 and DFDC remain external evaluation targets pending audited runs. Because the network is a binary classifier without a segmentation decoder, no pixel-level localization scores are reported; Grad-CAM and attention maps are qualitative feature-inspection tools only.

7 Discussion

The reported JPEG AUCs provide an initial measurement of image-level performance on the FFHQ-versus-StyleGAN2 collection: AUC is 0.937 at $Q = 50$ and 0.888 at $Q = 23$. The decline is consistent with increasing degradation, but the limited reported points and absence of uncertainty statistics limit the strength of any robustness claim. In particular, JPEG quality factors must not be relabelled as FF++ C23/C40, and image-level results should not be presented as video-benchmark results.

The reported architecture and WPD tests passed on an RTX 4060 Laptop GPU using PyTorch 2.6.0+cu124 and CUDA 12.4. These tests support the stated tensor shapes and differentiable implementation. They do not show that the full model is more accurate than the ablations, or that DCA prevents frequency-branch undertraining. Such conclusions require matched training runs, per-variant metrics, uncertainty estimates, and appropriate statistical tests.

The study defines five hypotheses covering augmentation (H1), incremental WPD contribution (H2), HiFE contribution (H3), DCA versus concatenation (H4), and domain-shift performance

(H5). The available measurements provide initial evidence for image-level JPEG evaluation and implementation correctness, but are insufficient to confirm H2–H5. The qualitative cross-generator observation should not be generalized to all synthesis families.

Limitations include the single active image dataset for the reported JPEG values, missing numerical results for JPEG Q=10 and Q=100 in the available experiment record, absence of complete matched-ablation and multi-seed tables, lack of fully specified numeric external-transfer metrics, and unavailable FF++ video data in the workspace described by the report. The architecture is an image-level binary classifier and does not provide pixel-level manipulation masks. JPEG recompression also cannot represent every social-media or video-transcoding pathway.

8 Conclusion

This paper documents a dual-branch deepfake image classifier built from an ImageNet-pretrained Xception spatial stream, a differentiable three-level db4 wavelet-packet branch, a learnable high-frequency enhancement module, and bidirectional dual cross-attention fusion. According to the implementation record, the WPD and architecture test suites passed on an NVIDIA RTX 4060 Laptop GPU under PyTorch 2.6.0+cu124 and CUDA 12.4. On the reported FFHQ-versus-StyleGAN2 image evaluation, ROC-AUC point estimates were 0.937 at JPEG quality 50 and 0.888 at JPEG quality 23; a decline at quality 10 was reported without a supplied numeric value. These findings are specific to the stated image dataset and JPEG protocol. Complete matched-ablation metrics, repeated-seed uncertainty, and fully quantified external-generator results are still needed before claiming that the proposed architecture components independently improve compression robustness or generalization.

Reproducibility and Data Statement

The reported verification environment is an NVIDIA GeForce RTX 4060 Laptop GPU (8 GB VRAM), PyTorch 2.6.0+cu124, and CUDA 12.4. The reported test outcomes are 13/13 architecture tests and 6/6 differentiable-WPD tests passing. To make the study independently reproducible, a submission package should include a stable repository URL, commit hash, dependency lock, dataset-manifest identifiers, training configurations, checkpoint identifiers, and scripts used to generate Table 3. Dataset content should only be redistributed where licensing permits. The completed manuscript must also include the target journal's required data-availability, funding, conflicts-of-interest, author-contribution, and AI-use declarations.

References

[1] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "FaceForensics++: Learning to detect manipulated facial images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 1–11.

[2] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 1251–1258, doi: 10.1109/CVPR.2017.195.

[3] J. Frank, T. Eisenhofer, L. Schönherr, A. Fischer, D. Kolossa, and T. Holz, "Leveraging frequency analysis for deep fake image recognition," in *Proc. 37th Int. Conf. Mach. Learn.*, vol. 119, 2020, pp. 3247–3258.

[4] Y. Qian, G. Yin, L. Sheng, Z. Chen, and J. Shao, "Thinking in frequency: Face forgery detection by mining frequency-aware clues," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 86–103.

[5] H. Kashiani, N. A. Talemi, and F. Afghah, "FreqDebias: Towards generalizable deepfake detection via consistency-driven frequency debiasing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 8775–8785, doi: 10.1109/CVPR52734.2025.00820.

[6] M. Wolter, F. Blanke, R. Heese, and J. Garcke, "Wavelet-packets for deepfake image analysis and detection," *Mach. Learn.*, vol. 111, pp. 4295–4327, 2022, doi: 10.1007/s10994-022-06225-5.

[7] S. Mallat, "A theory for multiresolution signal decomposition: The wavelet representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 11, no. 7, pp. 674–693, 1989.

[8] R. R. Coifman and M. V. Wickerhauser, "Entropy-based algorithms for best basis selection," *IEEE Trans. Inf. Theory*, vol. 38, no. 2, pp. 713–718, 1992.

[9] Y. Wang, K. Yu, C. Chen, X. Hu, and S. Peng, "Dynamic graph learning with content-guided spatial-frequency relation reasoning for deepfake detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 7278–7287.

[10] F. Wang, Q. Chen, B. Jing, Y. Tang, Z. Song, and B. Wang, "Deepfake detection based on the adaptive fusion of spatial-frequency features," *Int. J. Intell. Syst.*, vol. 2024, Art. no. 7578036, 2024, doi: 10.1155/2024/7578036.

[11] J. Gao, Z. Xia, G. L. Marcialis, C. Dang, J. Dai, and X. Feng, "Deepfake detection based on high-frequency enhancement network for highly compressed content," *Expert Syst. Appl.*, vol. 249, Art. no. 123732, 2024, doi: 10.1016/j.eswa.2024.123732.

[12] T. Ghosh and R. Naskar, "WGN: Wavelet-guided network for efficient and generalised deepfake detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2026, pp. 10984–10993.

[13] N. A. Chandra *et al.*, "Deepfake-Eval-2024: A multi-modal in-the-wild benchmark of deepfakes circulated in 2024," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2026, pp. 10668–10678.

[14] A. Vaswani *et al.*, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[15] T. Karras, S. Laine, M. Aittala, J. Hellsten, J. Lehtinen, and T. Aila, "Analyzing and improving the image quality of StyleGAN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 8110–8119.

[16] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-DF: A large-scale challenging dataset for deepfake forensics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 3207–3216, doi: 10.1109/CVPR42600.2020.00327.

[17] B. Dolhansky, J. Bitton, B. Pflaum, J. Lu, R. Howes, M. Wang, and C. C. Ferrer, "The DeepFake Detection Challenge (DFDC) dataset," arXiv:2006.07397, 2020.